

Math 526, S12

Name:

Quiz 13

Section:

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (6 points) Find an orthogonal matrix $\mathbf{Q}$ that diagonalizes $\mathbf{A} = \begin{bmatrix} -2 & 6 \\ 6 & 7 \end{bmatrix}$, i.e., $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$.

Problem 2. (4 points) For what numbers c is $\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & c & 1 \\ 0 & 1 & c \end{bmatrix}$ positive definite?